

Problem 7

(a) We apply Fact 1 from the handout, with $X_1 \leftrightarrow X$, $X_2 \leftrightarrow Y$, $x_2 = y$.

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} \bar{x} \\ \bar{y} \end{bmatrix}$$

$$\text{and } \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} = \begin{bmatrix} P & PC^T \\ CP & CPC^T + Q \end{bmatrix}$$

Then ~~Recall~~ $X|Y=y$ has distribution

$$\begin{aligned} \mu_{1|2} &= \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2) \\ &= \bar{x} + PC^T [CPC^T + Q]^{-1} (y - \bar{y}) \end{aligned}$$

and covariance

$$\begin{aligned} \Sigma_{1|2} &= \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \\ &= P - PC^T [CPC^T + Q]^{-1} CP \end{aligned}$$

Comparing to Problem 3, we see
that $\hat{X} = \mu_{1|2} = E\{X|Y=y\}$ $\square$

~~all~~

(b). The Schur complement of Σ_{11}
in $\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$ is

$$\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}.$$

Identifying $\Sigma_{11} \leftrightarrow P$, $\Sigma_{12} \leftrightarrow PC^T$

$\Sigma_{21} \leftrightarrow CP$ and $\Sigma_{22} \leftrightarrow CPC^T + Q$

we see that the Schur complement of P

in Σ is

$$P - PC^T(CPC^T + Q)^{-1}CP$$

and this the covariance of the random
vector $X|Y=y$. conditional $\square$